BACHELOR OF SCIENCE WITH HONOURS IN COMPUTING SCIENCE

Final Year Project Proposal

Drive Buddy – a Machine Learning-Enhanced Drowsiness and Gaze Distraction Monitoring System

Report by

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Date

February 2024

DECLARATION STATEMENT

I certify that the work submitted is my own and that any material derived or quoted from the published or unpublished work of other persons has been duly acknowledged.

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ABSTRACT

Drive Buddy, is a co-pilot safety system employing Eye Aspect Ratio (EAR) Logic and Machine Learning to detect drowsiness through Eye Aspect Ratio analysis. Using Image Recognition system, the solution aims detect fatigue markers as well other eye distractions. This study provide details of the conceptual design and basic design for the development of this solution. The study reviews notable research works conducted in the area of Eye Aspect Ratio and how to leverage on EAR to measure fatigues. These research studies will layout the foundation of this development study and use the research concept, outcomes and challenges to develop a practical solution. Furthermore, the later part of this focuses on the “how” of the solution development will came to be, the methods to be use, the development plan, and timeline to bring the solution to fruition.

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GLOSSARY

Drive Buddy System - An advanced driver monitoring system designed to detect signs of driver fatigue and enhance road safety through real-time analysis of facial features and driving behavior.

Driver Fatigue Status - The calculation and display of a fatigue score based on multiple indicators, providing the driver with insights into their current fatigue level.

Driving Behavior Scoring - The assignment of points or scores based on positive and negative driving behaviors, offering feedback to encourage safer driving practices.

EAR – Eye Aspect Ratio. A facial feature measurement algorithm that calculates the ratio of distances between specific landmarks on the eyes. Used for eye blink detection and monitoring eye movement.

EAR Threshold - The predefined value used in the EAR algorithm to determine the threshold for blink detection. The threshold may be adjusted based on individual characteristics.

MAR - Mouth Aspect Ratio. A facial feature measurement algorithm focused on the ratio of distances between facial landmarks around the mouth. Applied for detecting yawning as an additional indicator of driver fatigue.

1. Introduction

1.1 Background

In today's rapidly evolving technological landscape, the integration of smart systems into various aspects of daily life has become increasingly prevalent. This project responds to the growing need for innovative solutions in the realm of driver safety. With the rise of smart vehicles and the ever-expanding demands on individuals in transit, ensuring a secure and attentive driving environment is paramount.

While there are numerous research studies conducted on Eye Aspect Ratio (EAR), employing both conventional methods and modern approaches using image recognition software, a notable gap remains in translating these findings into practical solutions. Despite the wealth of knowledge accumulated, there is a lack of tangible products available for consumers. Therefore, the intention of this project is to bridge this gap by leveraging the outputs of research studies to develop a practical and user-friendly product—introducing the proactive and intelligent co-pilot system known as "Drive Buddy."

1.2 Problem Statement

In the early morning of April 25, 1981, Avelino Dizon, a young executive at Pepsi, was on his way home after a late-night sales plant visit in Tarlac City, Philippines. With only 40 kilometers left in his journey, fatigue took its toll, causing him to drift into sleep behind the wheel. The car veered off course, fell into a ditch and colliding with a tree. Tragically, Avelino lost his life on the scene, leaving an indelible mark on the lives of my wife and her family. Avelino's story joins the grim statistics, underscoring the consequences of drowsy driving incidents.

According to Singapore Medical Journal, driver fatigue is one of the biggest health and safety concerns within the road transport sector (Lim & Chia, 2015). In 2017, US Department of Transportation National Highway Traffic Safety Administration (NHTSA) have estimated 91,000 police-reported crashes involved drowsy drivers. These crashes led to an estimated 50,000 people injured and nearly 800 deaths (NHTSA, 2021).

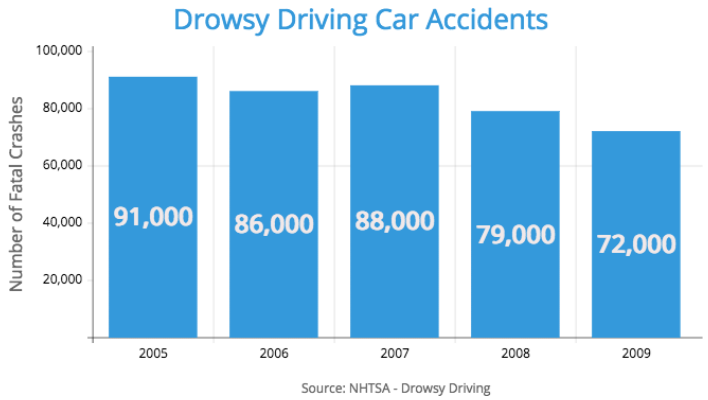


Figure 1 Statistics of Fatal Crashes due to Drowsy Driving (NHTSA, 2021)

The facts of the tragic incident and the statistics from the Singapore Medical Journal and NHTSA, underscore the urgent need for practical solutions to tackle drowsy driving. Taking a proactive stance is crucial to minimize the associated risks and prevent future accidents. It's imperative to develop and implement effective measures, like advanced co-pilot safety systems, to address this significant health and safety issue in the road transport sector. Transforming research findings into practical tools is key to fostering a safer driving environment, ultimately preventing loss of lives and sparing families from the enduring impacts of such incidents.

1.3 Objectives

The objective is this project is to develop an co-pilot safety system – the “Drive Buddy” incorporating Eye Aspect Ratio (EAR) Logic, machine learning, and intervention system for improved drowsiness detection and comprehensive fatigue management reducing accidents due to impaired driver alertness, enhancing road safety.

1.4 Scopes and Limitations

1.4.1 Scopes:

- Design, develop, and implement an co-pilot safety system named "Drive Buddy” using Internet of Things (IoT) solution;
- Incorporate Eye Aspect Ratio (EAR) Logic for real-time monitoring of driver alertness, using Open Computer Vision (OpenCV) to measure EAR.
- Utilize machine learning algorithms to enhance drowsiness detection capabilities.
- Implement an intervention system for fatigue management using Visual, Auditory, and Tactile methods.
- Develop a user-friendly mobile application for easy access and interaction and to gamify the driver driving behaviour.

1.4.2 Limitations:

- Effectiveness may be influenced by external factors like diverse driving conditions and individual user behaviours.
- Project scope excludes the physical manufacturing of hardware components, focusing on software and algorithmic development.
- The direct application of certain research outcomes from the EAR studies may have feasibility constraints.
- Despite efforts for continuous improvement, limitations identified during the project may persist in initial iterations.

2. Literation Review

2.1 Overview of Related Work

The referenced research paper, titled "Eye Aspect Ratio for Real-Time Drowsiness Detection to Improve Driver Safety," (Dewi, et al., 2022) addresses the critical issue of drowsiness as a major road safety risk. Investigating the intricate relationship between blink detection and driver safety, the study explores the challenges posed by the rapid blink rate and presents an innovative technique for real-time eye blink identification using a car dashboard camera.

The proposed method involves determining facial landmark positions and extracting the vertical distance between eyelids, utilizing the Eye Aspect Ratio (EAR). This algorithm efficiently estimates facial landmark positions, extracts a scalar quantity using EAR, and identifies eye closeness in each frame. The research establishes the optimal EAR threshold (0.18) through comprehensive experimentation, showcasing improved accuracy in detecting blinks.

The literature review highlights the significance of blink detection technology in various applications, including drowsiness detection, driver safety, computer vision, and face recognition systems. It underlines the importance of maintaining driver concentration during activities such as driving, which influences blink rates. The study also recognizes the impact of factors like eyelid conditions, eye conditions, and environmental stimuli on blink rates.

Several existing models and approaches are discussed, including those incorporating visual context patterns, hierarchical-based interpolation for fatigue indicators, and techniques involving eye closure for blink detection. The work emphasizes the integration of EAR and facial landmarks, drawing on successful methodologies to enhance blink detection accuracy.

2.2 Current State

The current state of eye blink detection involves a range of methods and techniques that utilize facial landmarks and algorithms for instantaneous analysis. It is essential to comprehend the existing literature to contextualize and acknowledge the contributions made by the present study.

2.2.1 Eye Aspect Ratio (EAR) Algorithm

The Eye Aspect Ratio algorithm, pioneered by Soukupova and Cech in 2016 (Soukupov´a, 2016), stands as a fundamental component in eye blink detection. By computing the ratio of distances between key facial landmarks, with a primary focus on the eyes, this algorithm proves highly effective in monitoring eye movement and blink patterns. Its simplicity and efficiency have led to widespread adoption in recent research endeavours.

2.2.2 EAR Threshold Variability

Explorations in prior research have delved into the influence of EAR threshold variability on the accuracy of blink detection (Dewi, et al., 2022). Despite the traditional preference for thresholds around 0.2, the current study challenges this norm. Through experimentation with various EAR thresholds (0.18, 0.2, 0.225, and 0.25), the study emphasizes the need to recalibrate thresholds to enhance accuracy, particularly in scenarios involving distinctive features such as small eyes.

2.2.3 Real-time Applications

In real-time applications, especially in domains like driver safety and drowsiness detection, accurate and prompt blink detection is important. Literature emphasizes that effective blink detection is crucial for averting accidents caused by driver drowsiness (Dewi, et al., 2022). The real-time capabilities demonstrated by the proposed method in the present study align with the increasing demand for efficient systems in practical applications.

2.3 Identified Gaps and Challenges

2.3.1 Gaps

- Conventional EAR thresholds around 0.2 were commonly used, but the study challenges this by experimenting with different thresholds (0.18, 0.2, 0.225, and 0.25).
- Real-time blink detection is crucial for applications like driver safety.
- Lack of details of denoising techniques for low-light situations.
- Ambient light level application of EAR thresholds

2.3.2 Challenges

- There's a need to recalibrate EAR thresholds based on individual characteristics like variations in eye size, challenging the one-size-fits-all approach and emphasizing adaptive thresholding.
- Developing real-time capabilities, optimizing algorithms for efficiency, and minimizing processing delays pose challenges.
- Getting the right thresholds for low-light scenarios pose a challenge to understand what’s the reasonable thresholds.
- Usage of the right equipment to capture image for low light scenario is another challenge.

2.4 Relevance to the Proposed Project

2.4.1 EAR Threshold Optimization

Relevance: The exploration of various EAR thresholds (0.18, 0.2, 0.225, and 0.25) directly informs our project, emphasizing the importance of fine-tuning the EAR threshold for accurate blink detection.

2.4.2 Individualized Adaptive Thresholding

Relevance: The call for adaptive thresholding based on individual characteristics aligns with our project's focus on tailoring the blink detection algorithm to diverse users, addressing challenges related to eye size variations.

2.4.3 Real-time Application Considerations

Relevance: The recognition of the importance of real-time capabilities in practical applications, especially in contexts like driver safety, underscores the significance of our project's goal to achieve efficient, prompt blink detection.

3. Project Description

3.1 System Architecture

Drive Buddy system integrates machine learning, IoT, and real-time analytics to combat driver fatigue and distractions for enhanced road safety. Leveraging on Edge ML to provide real-time inference based from EAR data collected by Raspberry Pi, HD Camera and OpenCV. The solution leverages on Amazon SageMaker platform to train and deploy models to SageMaker Neo (EdgeML) to predict driving fatigue and gaze distractions. React powers the User Interface Layer, offering a user-friendly mobile app for real-time insights.

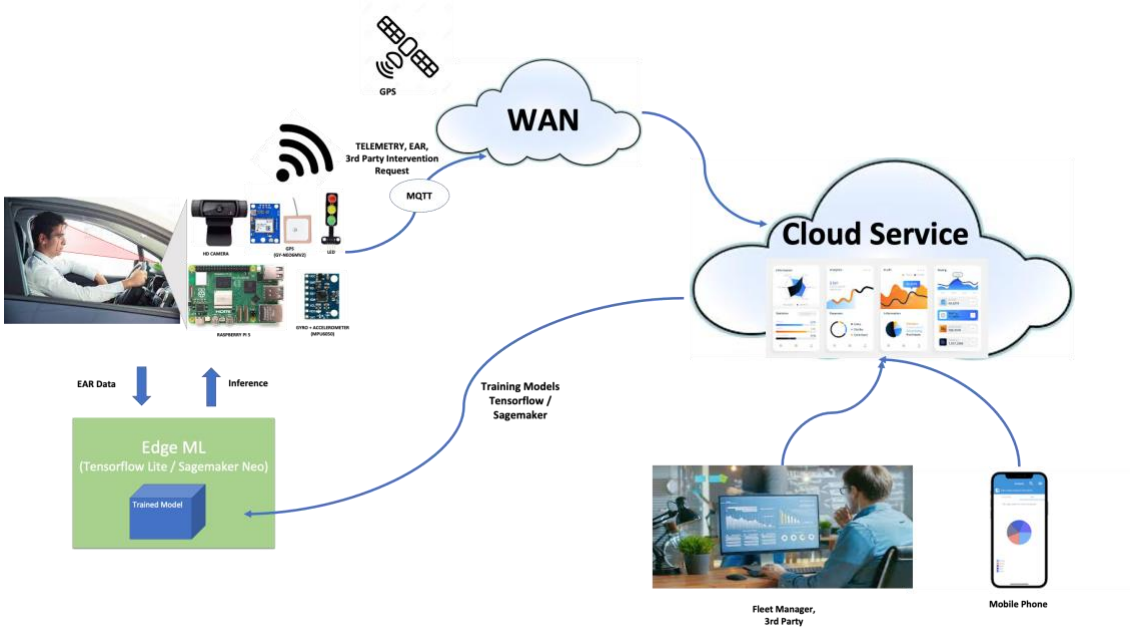


Figure 2 High Level System Architecture of Drive Buddy System

3.2 Technologies and Tools

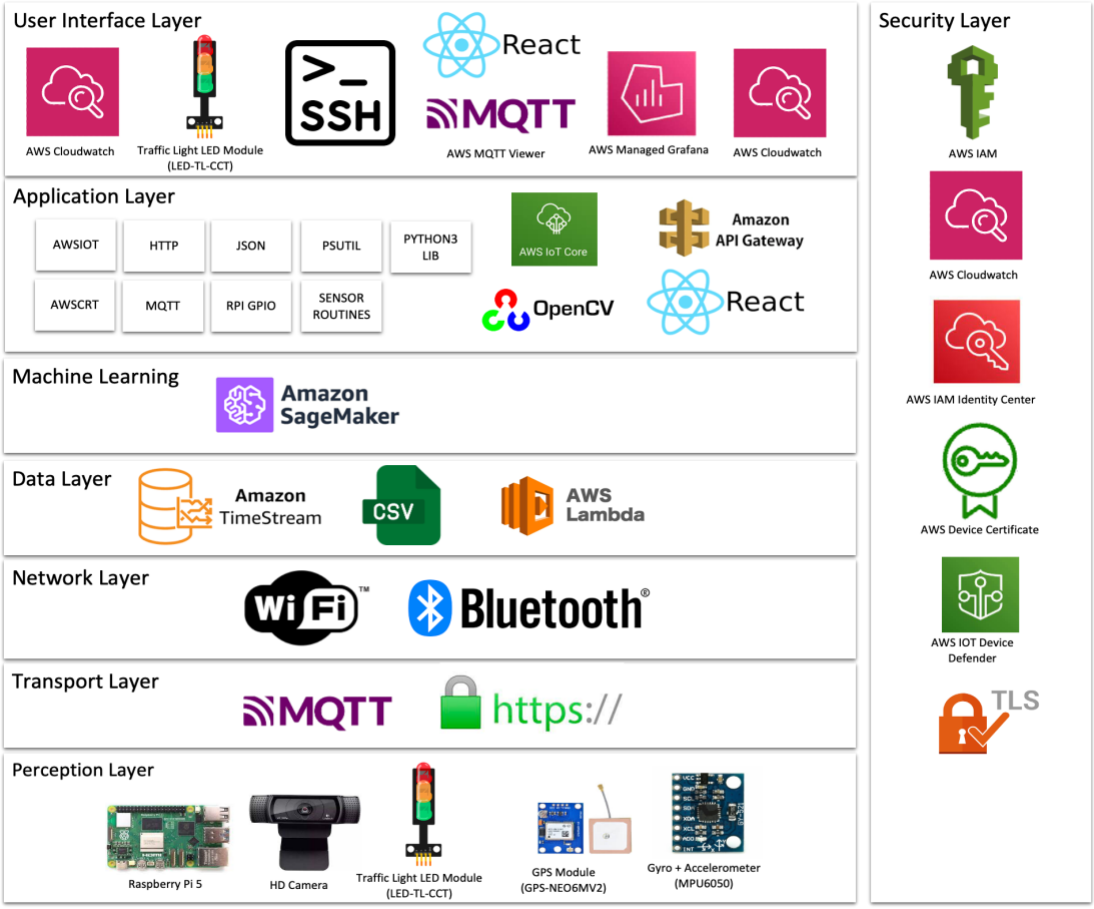


Figure 3 Layered Architecture of Drive Buddy with the Technology Used by the System

Table 1 Technology and Tools used by Drive Buddy System

Layer	Technology	Description
User Interface Layer	React	JavaScript library for building user interfaces
Security Layer	SSH	Secure shell protocol for remote access
	MQTT	Messaging protocol for machine-to-machine communication
Application Layer	Amazon API Gateway	Service that lets you manage APIs and connect them to your backend applications
	AWS IoT Core	A managed service that lets you connect and manage IoT devices
	PSUTIL	Cross-platform library for retrieving system information
	Python 3	General-purpose programming language
	AWS CloudWatch	Monitoring service that provides data and insights on your AWS resources and applications
Machine Learning Layer	Amazon SageMaker	Service that lets you build, train, and deploy machine learning models
Data Layer	Amazon TimeStream	Time series database service for IoT applications
	CSV	Comma-separated values file format
	AWS Lambda	Serverless compute service that lets you run code without provisioning or managing servers
Network Layer	Wi-Fi	Wireless networking technology
	Bluetooth®	Wireless technology for short-range data exchange
Transport Layer	MQTT	Messaging protocol for machine-to-machine communication
	HTTPS	Secure version of the HTTP protocol
Perception Layer	Raspberry Pi	Small, single-board computer
	HD Camera	High-definition camera
	Traffic Light LED Module	Module that controls traffic lights
	GPS Module	Module that provides location data
	Gyro + Accelerometer	Module that provides data on the orientation and motion of a device

3.3 Key Features and Functionalities

Table 2 Key Features and Functionalities of Drive Buddy System

#	Function	Description

1	Blink Metering	Measures eye blink frequency and duration to identify fatigue patterns.
2	Gaze Metering	Monitors gaze direction, head movements, and attention duration to detect prolonged diversions from the road.
3	Facial Gestures Detection (Yawning)	Identifies yawning by analyzing mouth shape and facial expressions.
4	Visual Alert System (Using LED)	Uses strategically placed LEDs to create visual cues that alert the driver.
5	Voice Intervention System (Using Pre-Worded Warnings)	Delivers audible warnings or reminders to encourage alertness.
6	Third-Party Notification Request	Sends notifications to designated contacts if drowsiness persists.
7	Face Recognition for Driver Logging	Identifies authorized drivers and personalizes monitoring parameters.
8	Vehicle Tracking System	Monitors vehicle location, speed, and other driving metrics.
9	Driver Fatigue Status	Calculates and displays a fatigue score based on multiple indicators.
10	Driving Behavior Score/Points	Assigns points or scores based on positive and negative driving behaviors.
11	Machine Learning Predictive Fatigue Analysis	Predicts driver fatigue using machine learning algorithms and historical data.

3.4 Stakeholders and User

3.5 Use Cases

Figure 4 Use Case Diagram of Drive Buddy System

3.6 Stakeholder and User

Table 3 Intended Stakeholders and Users

Role	Description	Responsibilities	Importance
Driver	Primary user	Operates the vehicle, interacts with alerts and interventions, monitors fatigue status.	Key to system effectiveness for road safety and preventing drowsiness-related accidents.
System	Drowsiness Detection System	Monitors driver behavior, analyzes sensor data, triggers alerts, manages functionality.	Accuracy and efficiency directly impact reliability of fatigue detection and prevention.
Fleet Manager	Oversees equipped vehicles	Analyzes fleet-wide data, accesses tracking information, ensures driver well-being and safety.	Leverages system for enhanced fleet management, driver safety, and performance evaluation.
Third-Party Contact	External party (family, emergency contact)	Receives notifications for persistent drowsiness, may trigger interventions.	Extends support network, enhancing driver safety beyond individual awareness.

4. Methodology

3.7 Research Methods

The study employs research methods that aim to systematically examine and assess the efficacy of the Drive Buddy system in mitigating driver fatigue and improving road safety. These methodologies integrate both quantitative and qualitative strategies, facilitating a thorough exploration of the system's operational effectiveness and user perceptions.

Table 4 To be adopted Research Methodologies

#	Phase	Description	Objective
1	System Implementation and Configuration	Implement Drive Buddy on diverse vehicles, optimize configuration parameters.	Ensure practicality and adaptability across driving conditions.
2	Data Collection	Gather data from sensors (eye-tracking, facial recognition, gyroscopes, accelerometers) in various driving scenarios.	Acquire diverse and representative data for model training and performance evaluation.
3	Machine Learning Model Training	Train models using collected data, optimizing parameters for accuracy and efficiency.	Develop robust models for accurate fatigue detection.

4	User Experience Evaluation	Conduct user study to assess driver experiences, feedback on alerts, usability, and satisfaction.	Understand user perspectives, identify usability issues, gather insights for improvement.
5	Comparative Analysis	Compare Drive Buddy's performance with existing solutions using key metrics (accuracy, false positives, response time).	Benchmark against industry standards, assess superiority in real-world applications.
6	Ethical Considerations	Adhere to ethical guidelines regarding privacy, data security, informed consent, and anonymization.	Ensure ethical conduct, protect rights and privacy of participants.

3.8 Data-Collection and Analysis

This section outlines the methodologies employed to collect relevant information and the subsequent analytical techniques applied to derive meaningful insights.

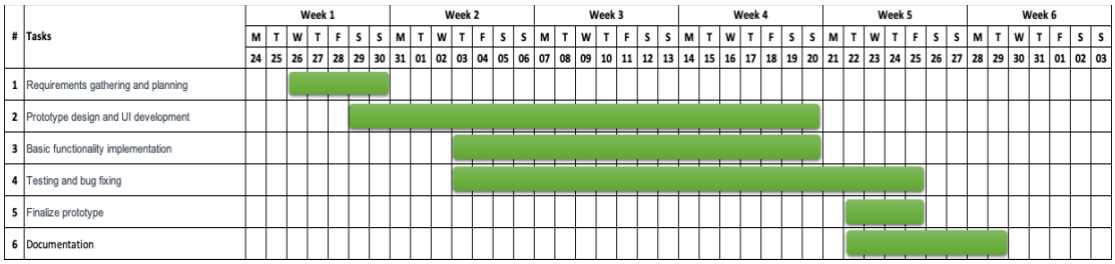
Table 5 Data Collection and Analysis to be adopted

Index	Method	Description	Objective	Data Source
1	Sensor-Based Data Collection	Collect data from eye-tracking, facial recognition, gyroscopes, accelerometers in diverse driving scenarios.	Gather comprehensive data for training models and evaluating system performance.	Real-world driving scenarios.
2	User Interaction Logging	Log responses to visual and voice alerts, instances of alert triggers, and driver reactions.	Understand driver interactions, identify response patterns, and assess alert effectiveness.	System usage logs.
3	Questionnaire and Interviews	Administer questionnaires and conduct interviews to gather qualitative feedback on user experience, alert effectiveness, and suggestions for improvement.	Obtain subjective insights to complement quantitative data.	Participating drivers.
4	Facial Recognition Data Anonymization	Anonymize and encrypt facial data for privacy protection.	Ensure compliance with privacy regulations and safeguard user identities.	Collected facial recognition data.

5	Machine Learning Model Evaluation	Assess model performance using metrics like accuracy, precision, recall, F1-score. Conduct cross-validation.	Ensure effectiveness of fatigue detection algorithms and identify areas for improvement.	Model evaluation results.
6	Driving Behavior Logging	Log speed, acceleration, deviations, and correlate with fatigue events.	Understand the relationship between driving behavior and fatigue, identifying potential precursors to drowsiness.	Driving behavior logs and fatigue events.

3.9 Project Management

The project's lifecycle is divided into key phases in 6 weeks period (time allocated for this project), each strategically planned to ensure a systematic and efficient progression. The Gantt chart below outlines the major milestones and tasks, providing a visual representation of the project's timeline and dependencies.



3.10 Development Approach

The proposed development approach involves a hybrid model, combining elements of both waterfall and agile methodologies. The initial phase will follow a waterfall structure, focusing on establishing the basic project structure and implementing key features. Once this foundation is in place, subsequent development will transition to an agile methodology, allowing for more flexibility and iterative enhancements. The overall project timeline is set for six weeks, starting on December 26th and concluding on February 3rd.

3.11 Testing and Validation

To ensure the robustness and reliability of the Drive Buddy system. This section outlines the comprehensive strategies employed to assess the functionality, performance, and security aspects of the system.

Table 6 Testing and Validation intended for Drive Buddy project

Section	Details
Testing and Validation Overview	Provides an overview of the testing and validation process for the Drive Buddy system.
Testing Objectives	Outlines the specific goals of testing, including:
	Functionality: Verifying core features like drowsiness detection, alerts, and third-party notifications.
	Performance: Evaluating system responsiveness under various conditions and optimizing resource utilization.
	Security: Assessing resilience to cyber threats and identifying potential vulnerabilities.

	Usability: Ensuring a user-friendly interface and positive user experience.
Validation Strategies	Describes the methods used to validate the system's effectiveness, including:
	Scenario-based Testing: Simulating real-world driving scenarios to assess performance.
	User Feedback Collection: Gathering feedback from drivers, administrators, and third-party contacts.
	Continuous Improvement: Implementing an iterative process for ongoing refinement based on feedback and evolving requirements.

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APPENDICES